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Occurrence of banana bunchy top virus in tissue-cultured and conventional banana cultivars in the Siang region of Arunachal Pradesh, India

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Banana bunchy top virus (BBTV; genus *Babuvirus*, family *Nanoviridae*) is one of the most destructive viral pathogens of banana (*Musa* spp.), causing severe stunting, bunching, and yield losses that may reach up to 100% in highly susceptible cultivars (Nelson and Thomas, 2020). The virus is transmitted persistently by the banana aphid *Pentalonia nigronervosa Coquerel* and through infected planting material, including tissue-cultured plants when proper virus indexing is not followed (Jebakumar *et al*., 2018). Although banana is an important fruit crop in Arunachal Pradesh, systematic information on BBTV occurrence in the Siang region is limited. The present study documents the occurrence, incidence, and molecular confirmation of BBTV in tissue-cultured and conventional banana cultivars in this region.

A systematic orchard survey was conducted during August-December 2025 in the Siang region of Arunachal Pradesh, covering Likabali (≈240 m AMSL), Aalo (≈300 m AMSL), and Basar (≈650 m AMSL). Banana plants were examined for characteristic BBTV symptoms, including marginal chlorosis, dark green discontinuous streaks ("Morse code") on the midrib and veins, J-shaped hooks, narrow upright leaves, bunching, and severe stunting. Disease incidence was calculated as the percentage of symptomatic plants relative to the total number of plants observed at each location.

Both tissue-cultured banana cv. Grand Nain and conventionally propagated cultivars (Dwarf Cavendish, Giant Cavendish, Monthan [Kas Kol], and Poovan) were surveyed. Wild *Musa* species (*M. rosacea* Jacq. and *M. velutina* Wendl.) occurring naturally in the region were also examined. Young leaf samples from symptomatic and asymptomatic plants were collected, and total genomic DNA was extracted using the CTAB method. PCR amplification targeting the BBTV DNA-S (coat protein) component was carried out following Manickam *et al.* (2002). Amplified products were resolved on 1.2% agarose gel and visualized under UV illumination. PCR amplification using BBTV DNA-R-specific primers was performed to further confirm BBTV infection. PCR amplification yielded the expected ~1.0 kb amplicon in all symptomatic samples, whereas asymptomatic plants and healthy controls showed no amplification, thereby confirming the presence of Banana bunchy top virus in the surveyed banana cultivars.

Characteristic BBTV symptoms were observed in several banana cultivars under natural orchard conditions. In tissue-cultured banana cv. Grand Nain (Fig. 1), disease incidence ranged from 13.0% to 22.0% across locations, with a pooled incidence of 17.66% (Table 1). Infected plants were predominantly 4-6 months old, indicating early establishment of infection. Among conventional cultivars, Dwarf Cavendish showed very high susceptibility (Fig. 2), with disease incidence reaching 95.0% at Likabali, followed by Basar (87.5%) and Aalo (62.5%). Giant Cavendish and Monthan (Kas Kol) exhibited moderate infection levels (37.5-45.0%), whereas Poovan remained completely free from symptoms. Wild *Musa* species (*M. rosacea* and *M. velutina*) also showed no symptoms and tested negative by PCR (Table 2).

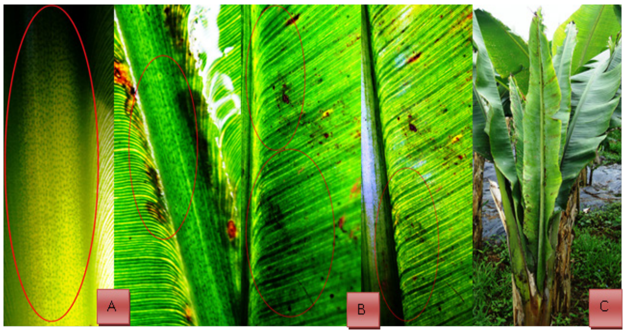

Figure 1. Symptoms of BBTV on tissue-cultured banana cv. Grand Nain: (A) Morse code streaking on leaf midrib; (B) J-shaped hooks along the midrib; (C) bunching symptom

Table 1. Incidence of BBTV in tissue-cultured banana (cv. Grand Nain) under natural orchard conditions in the Siang region, Arunachal Pradesh (Aug-Dec 2025)

Location	Total plants surveyed	Symptomatic plants	Disease incidence (%)	Age of infected plants (months)	Source of planting material	PCR confirmation
Likabali	100	22	22.0	4-6		Positive
Aalo	100	18	18.0	4-6	Commercial nurseries, Assam (India)	Positive
Basar	100	13	13.0	4-6		Positive
Total	300	53	17.66			

Table 2. Reaction of banana cultivars and wild *Musa* species against BBTV under natural orchard conditions in the Siang region, Arunachal Pradesh

Variety / Cultivar	Location	Total plants observed	Symptomatic plants	Disease incidence (%)	Age of infected plants (months)	Source of planting material
Dwarf Cavendish	Likabali	40	38	95.0	5-6	
	Aalo	40	25	62.5	5-6	Commercial nurseries, Assam (India)
	Basar	40	35	87.5	5-6	
Giant Cavendish	Basar	40	15	37.5	3-5	
Monthan (Kas Kol)	Basar	40	18	45.0	4-6	
Poovan	Basar	40	0	0.0	5-6	
Musa rosacea Jacq.	Likabali	40	0	0.0	6-7	Wild sources
Musa velutina Wendl.		40	0	0.0	6-7	Wild sources

PCR amplification of BBTV DNA-S and DNA-R genomic components consistently produced the expected amplicons in symptomatic samples, while asymptomatic plants remained negative (Fig. 2), confirming BBTV infection. The detection of BBTV in tissue-cultured Grand Nain highlights the risk associated with inadequate virus indexing of planting material, whereas the extremely high incidence in Dwarf Cavendish reflects widespread exposure and efficient aphid-mediated transmission. The absence of infection in Poovan and wild *Musa* species suggests possible resistance or escape under natural orchard conditions, as reported earlier for certain banana genotypes (Nelson and Thomas, 2020). This study confirms the occurrence of Banana bunchy top virus in both tissue-cultured and conventional banana cultivars in the Siang region of Arunachal Pradesh. The findings emphasize the urgent need for the use of certified virus-free planting material, routine PCR-based surveillance, effective aphid management, and prompt removal of infected plants to prevent further spread and safeguard banana production in the region.

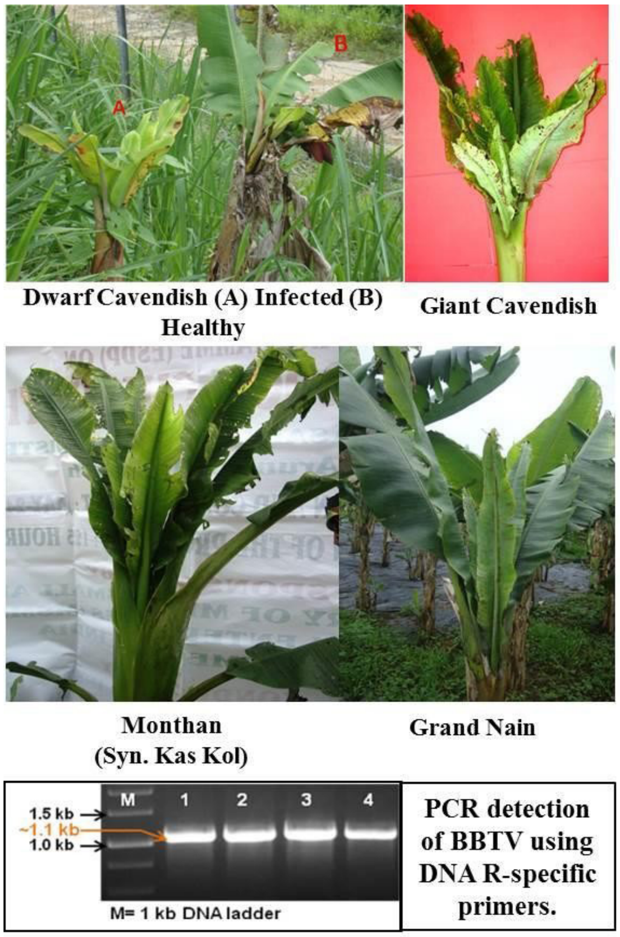

Figure 2. Severe BBTV symptoms on Dwarf Cavendish, Giant Cavendish, Monthan and Grand Nain and PCR-based detection of BBTV using DNA-S and DNA-R-specific primers. Lanes 1-4: symptomatic samples; M: 1 kb DNA ladder; expected amplicons (~1.1 kb for DNA-S and ~1.0 kb for DNA-R)

Acknowledgment

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